Data Transformation to Handle Skewness in Machine Learning

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1. Introduction

Skewness is a statistical measure that describes the asymmetry of a probability distribution around its mean. In essence, it tells us which direction our data is leaning toward.

2. Understanding Skewness

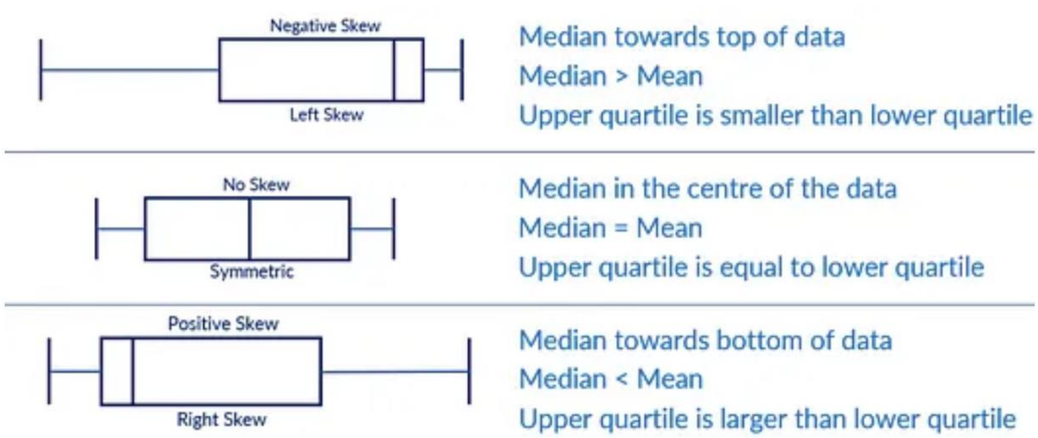

Figure 1. Negative skew, symmetric distribution, and positive skew, showing the relative positions of the mean, median, and mode.

Type	Tail Direction	Mean, Median, Mode Relationship	Real-World Example
Symmetric	Equal on both sides	Mean = Median = Mode	Human height in a population
Positive skew (+ve)	Extends to the right	Mean > Median > Mode	Household prices
Negative skew (–ve)	Extends to the left	Mean < Median < Mode	Age at death

Skewness tells us how our data leans, giving us a sense of which values dominate the dataset. When skewness is positive (right-skewed), most of the data is concentrated at the lower end, with outliers at the higher end pulling the mean to the right. When skewness is negative (left-skewed), most of the data is concentrated at the higher end, with outliers at the lower end pulling the mean to the left.

It follows the formula $\gamma_1 = E[((X - \mu) / \sigma)^3]$, where $X - \mu$ is the distance from the mean and σ is the standard deviation.

3. Why Skewness Can Be Harmful

3.1 Distorts Statistics

When data is skewed and the mean is pulled toward the tail, imputing missing values with the mean becomes inaccurate, since the mean is higher or lower than most of the data, which introduces bias. Standard deviation and variance also rely on the mean, so any distortion in the mean distorts the standard deviation and, in turn, the z-score.

3.2 Warps Distance-Based Algorithms

Algorithms that compute distances between points are highly sensitive to extreme tails.

KNN and K-Means models: extreme tail values dominate distance calculations. This forces clusters to stretch toward outliers or causes normal points to be misclassified.

3.3 Violates Assumptions of Linear and Parametric Models

Many classic algorithms assume that features or residual errors follow a normal distribution.

Linear and logistic regression: due to skewness, these models make far larger prediction errors on extreme tail values than on typical values.

4. Detecting Skewness

There are a few ways to detect skewness, such as box plots, which rely on the interquartile range (IQR):

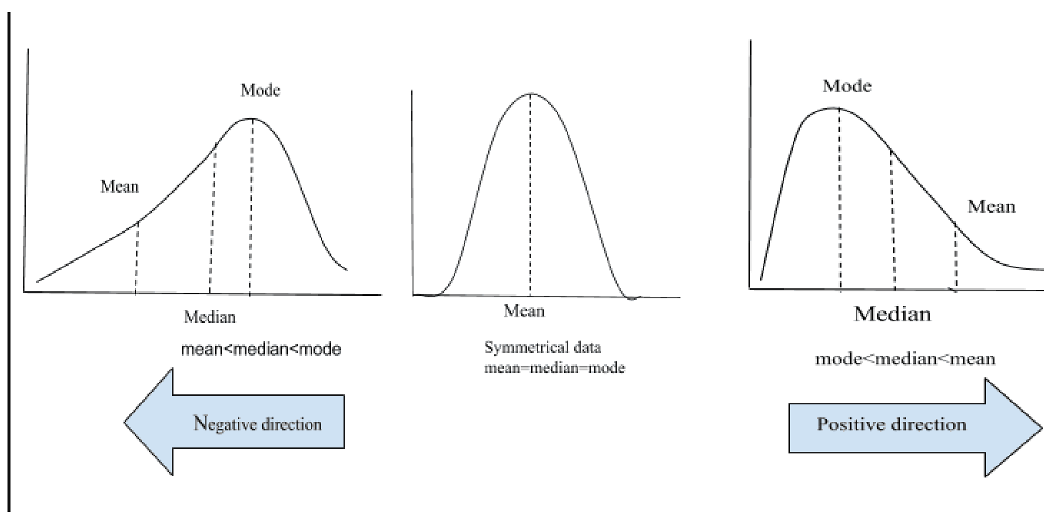

Figure 2. Box plot patterns for negative skew, no skew, and positive skew, and how each relates to the median-mean relationship and quartile spread.

...and histograms with a KDE (kernel density estimation) overlay:

Figure 3. A right-skewed (positively skewed) histogram with its KDE curve overlaid.

The blue line represents the KDE, which is essentially an estimate of the shape of the distribution curve for the given data.

5. Handling Positive Skewness

As established, skewness must be handled for gradient- and distance-based models to work correctly. Several methods exist to compress the tail and bring the data closer to a normal distribution.

5.1 Log Transformation

Compresses higher values much more than lower values. Best suited for severe right skewness spanning several orders of magnitude. Cannot handle negative values.

5.2 Square Root Transformation

Applies mild compression to high values. Best for mildly skewed data; cannot handle negative values.

5.3 Cubic Root Transformation

Similar to the square root transformation but with slightly stronger compression. Can handle negative values naturally, and works best with moderately skewed data containing negative values, without needing a constant shift.

5.4 Reciprocal Transformation

Inverts the scale: small numbers become extremely large, while large numbers become extremely small. Its drawbacks are that it reverses the overall order of the data and cannot handle zero values.

5.5 Box-Cox Transformation

$y^{(\lambda)} = \ln(y)$, for $y = 0$; $(y^{\lambda} - 1) / \lambda$, for $y \neq 0$

It selects a lambda value for positive data, using an optimization algorithm (maximum likelihood estimation) to estimate the lambda that yields the greatest normalization. It cannot handle negative values.

5.6 Yeo-Johnson Transformation

The most widely used algorithm for handling skewed data. It can handle negative, positive, and zero values, and is an extension of the Box-Cox transformation.

6. Handling Negative Skewness

Negative skewness occurs when a dataset has a long tail on the left side, extending toward smaller values, with most of the data concentrated on the right. Standard algorithms like Box-Cox are designed for right-skewed data and work by compressing large values, so applying them to negatively skewed data only makes the problem worse.

Rather than creating a separate algorithm for negative data, the dataset can be reflected, or mirrored, along the y-axis. Once reflected, the data behaves like a positive skew, and standard algorithms can be applied.

Yeo-Johnson can handle negative data directly, without the need for mirroring or reflection, since it natively supports negative values. It is generally preferred, as it requires no additional preprocessing.

7. Box-Cox vs. Yeo-Johnson Comparison

Feature / Metric	Box-Cox	Yeo-Johnson
Data type	Only works with positive data	Works with all types of data
Advantages	Mathematically simpler. Highly effective for right-skewed positive datasets. Straightforward parameter estimation and inversion.	Handles zeros and negative numbers without constant offsets or reflections. More versatile for modern machine learning pipelines where features are standardized or centered around zero.
Disadvantages	Requires a constant shift or offset to handle negative values. Requires reflection to handle left-skewed data.	Slightly more complex math. Can still be sensitive to extreme outliers, which may bias parameter estimation. Interpretability of transformed units remains abstract (requires an inverse transformation for physical meaning).
Suitable use cases	Positive physical measurement data. Financial metrics bound above zero (house prices, income).	Normalized, mean-centered, or standardized datasets with mixed positive and negative values. Datasets containing zeros. Feature engineering in scikit-learn pipelines.

8. Practical Example

While exploring the housing.csv file, I found that most of the data was right-skewed.

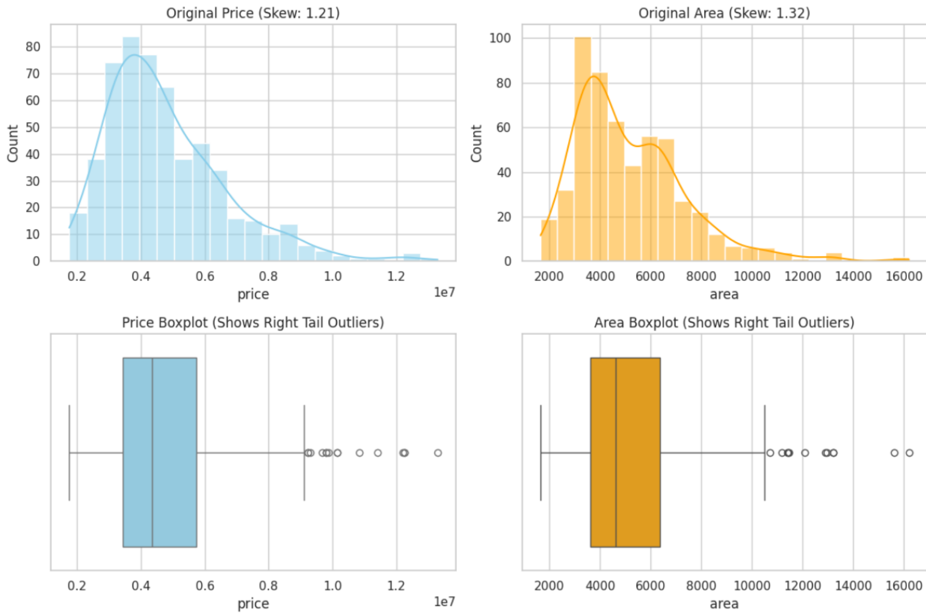

Figure 4. Original distributions of house price and area, with histograms (top) showing positive skew and box plots (bottom) showing right-tail outliers.

The prices and areas of the houses were positively skewed, and the outliers are visible in the box plots, so the Yeo-Johnson transformation was applied.

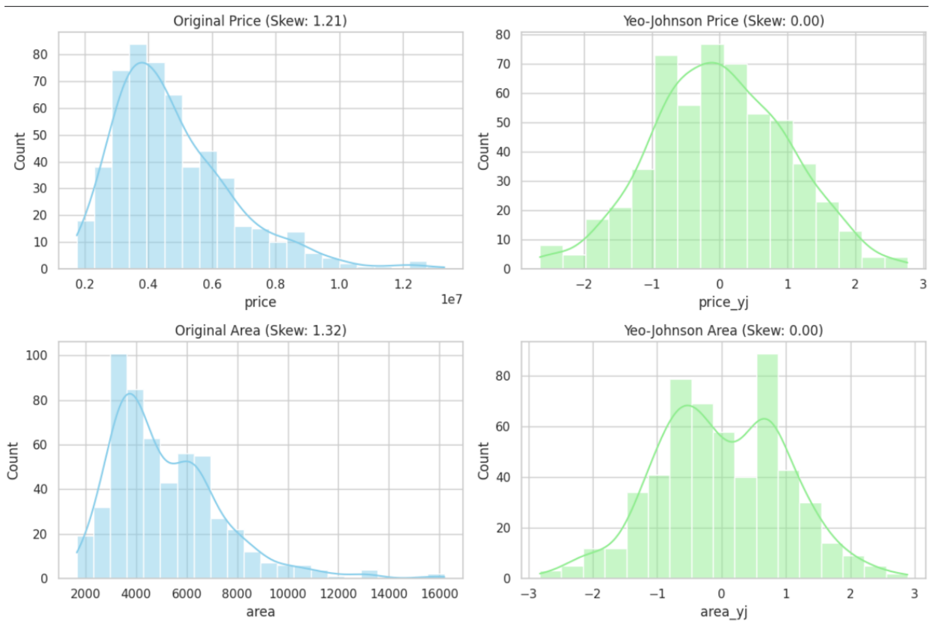

Figure 5. Price and area distributions before and after the Yeo-Johnson transformation, with skewness reduced to approximately zero.

These are the plots before and after the transformation, and the data is clearly much closer to a normal distribution now.

9. Conclusion

Data transformation is vital for data processing and for machine learning models to function correctly. Handling skewness stabilizes variance, neutralizes the effect of outliers, and ensures that distance-based and gradient-based models work as intended. While traditional transformation functions and Box-Cox are effective for certain data types, modern pipelines increasingly favor Yeo-Johnson for its versatility.